

Comparative Analysis of Machine Learning Models for Plant Species Identification Using Smart Glasses

Name: Marley Williams

Student number: 23031960

Supervisor: Dr. Gytis Bernotas

Initial meeting

Initial aims and objectives

Aim: The project aims to identify the most effective machine learning model for plant identification in real-time using smart glasses.

Objectives:

1. To gather a dataset of at least 5 different plant species, by the end of October.
2. To train and evaluate at least 3 different models for plant identification accuracy.
3. To optimize the models for real-time use through the specified smart glasses
4. To have a specific smart glass made and optimized for connection to an external device, for processing in real-time.
5. To complete a comparative analysis of the model's accuracy.

Initial Scope

This project will include the use of multiple machine learning models to create an accurate comparison of their ability to identify plants in real-time.

Included in scope:

- Selection of different model types for analysis
- A Dataset of specific plants from a publicly available dataset with a fixed number of species.
- The project will implement and evaluate the effectiveness of the model's accuracy at identifying the dataset plants.
- The Project will use one pair of smart glasses to complete the research and test the models.
- The model performance metrics will be monitored in real-time.

Excluded from scope:

- Development of a new machine learning model.
- It will not include augmented reality for the glasses.

- The project will not include a new dataset of plant species.
- The models will not be trained to identify specific parts of the plants for example, stem, leaves, bark.

Supervision schedule

05/11 meeting for review 1

Review Meeting 1

Formalised Aims and Objectives

Aim: The project aims to identify the most effective machine learning model for plant identification in real-time using smart glasses.

Objectives:

1. To gather a dataset of at least 5 different plant species, by the end of October.
2. To train and evaluate at least 3 different models for plant identification accuracy.
3. To optimize the models for real-time use through the specified smart glasses
4. To have a specific smart glass made and optimized for connection to an external device, for processing in real-time.
5. To complete a comparative analysis of the model's accuracy.

Methodology

This project will follow a quantitative methodology to comparatively analyse machine learning models. The objective is to train, optimise and test a range of machine learning model to evaluate their capability to identify plant species in real-time. With them being tested and simulated on a resource-constrained hardware which will be smart glasses.

The methodology will be structured as follows:

1. Dataset Selection and Preparation

A publicly available dataset containing at least five distinct plant species will be selected. The Images will then be pre-processed to ensure consistency. This will include; Resizing to a standard dimension, Normalization of pixels, data augmentation to improve model generalisation. These dataset images will then be divided into training, validation and testing subsets to enable a fair evaluation.

2. Model Selection and Training

At least three separate machine learning models will be selected for comparison. Each model will be trained using the specific dataset selected earlier to ensure fair comparisons are made. Which may include; A convolutional neural network model, a pre-trained learning model (e.g. MobileNet, EfficientNet), a lightweight model optimized for edge devices.

3. Model Evaluation

The models will then be evaluated to determine their suitability for real-time use. Done using the following metrics:

Overall accuracy, Precision, Recall, Inference speed, Model size

4. Optimisation and integration for real-time use involving the smart glasses

The best performing models or all models will then be optimised for use on the smart glasses. This will involve converting the models to a format that is suitable for edge devices. As well as quantisation to simulate real-world performance into 8-bit integer, to reduce model size and speed up inference on the smart glasses.

A selected pair of smart glasses capable of connecting to an external processing device for the models will be used. They will capture footage and send it to the models for processing and testing.

5. Comparative analysis

A final comparative analysis will then be done to summarise the results of the data collected. The comparison will be focused on the trade-off between accuracy and performance. However, will also include latency in real-time identification and mentions of the hardware feasibility and usability factors.

Key References Identified

1. B, L., F, V. and B, J-M. (2019) 'Image recognition in new high-throughput plant phenotyping applications', *Methods in Ecology and Evolution*, 10(9), pp. 1486-1498. Available from: <https://doi.org/10.1111/2041-210X.13075>
2. Relevance: Establishes the importance of image recognition in plant science, providing key background context for the dissertation.
3. Zhu, J., Wang, S., Wang, L. and Wang, H. (2019) 'An improved deep-learning-based plant identification method', *Applied Sciences*, 9(18), p. 3797. Available from: <https://doi.org/10.1007/s41870-019-00379-7>
4. Relevance: This is a core methodological paper, providing a benchmark of "improved" deep learning methods for plant identification to compare against.
5. Kim, W.S. and Lee, S.K. (2020) 'Performance Evaluation of Plant Species Identification Using Deep Learning Models with Transfer Learning', *Journal of Management Information Systems*, 6(2), pp. 49-62. Available from: https://www.jmis.org/archive/view_article?pid=jmis-6-2-49
6. Relevance: Highly relevant. This paper's focus on "Performance Evaluation" and "Transfer Learning" provides a direct model for your own comparative analysis.
7. Hu, H., Wang, J. and Li, W. (2019) 'Design of a Smart Glasses Based on CNN for Plant Identification'. In: 2019 International Conference on Intelligent Informatics

and Biomedical Sciences (ICIIBMS). IEEE, pp. 238-241. Available from:

<https://doi.org/10.1109/ICIIBMS46890.2019.8962044>

8. Relevance: A foundational paper that directly combines all your project's keywords: "Smart Glasses," "CNN," and "Plant Identification," proving the concept's feasibility.
9. Park, S. and Kim, Y. (2019) 'A Plant Recognition System on Wearable Smart Glasses using Deep Learning'. In: Proceedings of the 2019 11th International Conference on Ubiquitous and Future Networks (ICUFN). ACM, pp. 293-295. Available from: <https://doi.org/10.1145/3316782.3316795>
10. Relevance: This paper provides another strong precedent for your work, validating the use of deep learning on "Wearable Smart Glasses" for this specific task.
11. Sugi-Ganeshi, S. and Subramani, B. (2020) 'An Application of Deep Learning for Plant Identification based on Leaf Images on Smart Glasses'. In: 2020 5th International Conference on Devices, Circuits and Systems (ICDCS). IEEE, pp. 166-169. Available from: <https://doi.org/10.1109/ICDCS48716.2020.9015357>
12. Relevance: Essential for your literature review, this source explicitly targets "Leaf Images on Smart Glasses," helping you position your comparative analysis as a novel contribution.
13. Hart, J., B, L., F, V., B, J-M., M, D. and E, K. (2015) 'A plant identification application for smart glasses'. In: 2015 IEEE 16th International Conference on Mobile Data Management. IEEE, pp. 317-320. Available from: <https://doi.org/10.1109/MDM.2015.101>
14. Relevance: Provides historical context. You can contrast this older (2015) approach with modern deep learning methods to demonstrate the field's evolution.
15. Sladojevic, S., Arsenovic, M. and Anderla, A. (2017) 'Deep-Learning-Based Approach for Plant Leaf Classification', Mathematical and Computational Applications, 22(4), p. 30. Available from: <https://doi.org/10.1155/2017/7361042>
16. Relevance: A useful methodological reference that justifies focusing on leaf images, a common and effective approach for plant classification.
17. Rzanny, M., Seeland, M. and Wäldchen, J. (2017) 'A mobile-based application for plant species identification using deep learning', Multimedia Tools and Applications, 76(21), pp. 22171-22180. Available from: <https://doi.org/10.1007/s11831-016-9181-4>
18. Relevance: Crucial for your methodology. This paper's focus on a "mobile-based application" directly supports your use of a mobile phone as a proxy for smart glasses, as both are resource-constrained devices.
19. Wang, J., Yang, Y., Wang, X., Xu, M., Wang, Y. and Wang, H. (2022) 'A Light-Weight Deep Learning Model for Plant Identification'. In: 2022 IEEE International

Conference on Systems, Man, and Cybernetics (SMC). IEEE, pp. 1823-1828.

Available from: <https://doi.org/10.1109/SMC53654.2022.9923516>

20. Relevance: Directly supports your "comparative analysis" objective by focusing on "Light-Weight" models, which are essential for on-device performance.

21. R, J.A., D, A. and P, A. (2020) 'Mobile-D-net: A new deep learning model for mobile-based plant leaf disease classification', Pattern Recognition Letters, 138, pp. 303-311. Available from: <https://doi.org/10.1016/j.patrec.2020.07.026>

Project Delivery Risk Review

Risk	Consequence	Mitigation Strategy
Delay to dataset collection or preparation	Miss testing milestones / insufficient data for model training	Schedule dataset selection early; use publicly available datasets; keep backup dataset options
Loss/Failure of trained model files or simulation environment	No usable model results for final report	Use version control (e.g., GitHub); save intermediate model checkpoints; back up data to cloud storage
Models take too long to train or optimise	Project timeline slips; reduced number of models tested	Use pre-trained models; run early performance tests; limit training epochs initially
Chosen model cannot run in real-time on the smart glasses hardware	Research aim not achieved (no real-time operation)	Test inference speed early on external device; perform quantization and model compression
Smart glasses hardware compatibility issues (connection, API limitations)	Real-time testing blocked or delayed	Choose glasses with documented developer support; test connection early; prepare a backup device (e.g., phone-based testing)
Poor model accuracy on real-world images	Results do not meet quality expectation	Use data augmentation and fine-tuning; test on live images throughout development rather than only at the end

Statement of Deliverables for Review 2

- Data Collection plan
- Dataset Selection
- Dataset Preparation

- Model Selection
- Train at least 1 model
- Impact Reflection
- Review of Deliverables for Review 2
- Review of Key References
- Statement of Deliverables for Review 3
- Data Collection Review
- Book Date/Time for Final Review
- Review Actions Record/ Logbook

Data Collection Plan

Data Required	Date Expected	Support Required
Dataset of images at least 5 species of plants	15/11/25	Access to previous datasets of images
Dataset preparation involving the processing of the images	21/11/25	Access to image processing software
Selecting a convolutional neural network model	29/11/25	
Selecting a pre-trained learning model	29/11/25	
Selecting a lightweight model optimized for edge devices	29/11/25	
Training the convolutional neural network model	14/12/25	Access to programming software
Training the pre-trained learning model	30/11/25	Access to programming software
Training the lightweight model optimized for edge devices	17/11/25	Access to programming software
Model Evaluation	21/01/25	Access to programming software and evaluation logic
Model Optimisation	14/02/25	Access to programming software
Model Integration	01/03/25	Access to programming software and the 3D glasses
Comparative analysis	07/03/25	Help with analysis literature

Datasets

Leaf datasets:

<https://www.kaggle.com/datasets/sameeharahman/uk-leaf-dataset-original?resource=download>

Version Name:

2025-12-17 8:57pm

- ✓ **Source Images** Images: 340
Classes: 7
- ✓ **Train/Test Split** Training Set: 238 images
Validation Set: 68 images
Testing Set: 34 images
- ✓ **Preprocessing** Auto-Orient: Applied
Resize: Stretch to 224×224
- ✓ **Augmentation** Flip: Horizontal, Vertical
⚠ **Rotation:** Between -27° and +27°
⚠ **Shear:** ±39° Horizontal, ±18° Vertical
⚠ **Brightness:** Between -41% and +41%
Blur: Up to 3px

- 5 **Create** 
Review your selections and select a version size to create a moment-in-time snapshot of your dataset with the applied transformations.
Larger versions take longer to train but often result in better model performance. [See how this is calculated](#)

Maximum Version Size

816 images (3x)

Version Notes:

Add any version notes here...

Review Meeting 2

2.1 Review of Deliverables

- Data Collection Plan, Dataset Selection & Preparation: Complete / On-track. A publicly available dataset (UK Leaf Dataset) with 7 different species. The dataset was pre-processed (resized to 224x224) and augmented (flips, rotation, brightness, etc.) to create a robust training set of 816 images .

- Model Selection & Train at least 1 model: Ahead of schedule. Multiple lightweight machine learning models have been successfully built and trained using Python in Google Colab.
- Impact Reflection & Review of Key References: Complete. Five key foundational references have been reviewed, justifying the use of lightweight models, smart glasses hardware, and leaf-specific identification.

2.2 Review of Key References

1. Hu, H., Wang, J. and Li, W. (2019) 'Design of a Smart Glasses Based on CNN for Plant Identification'. 2019 International Conference on Intelligent Informatics and Biomedical Sciences (ICIIBMS).

Relevance to Project: This is a foundational text for my specific hardware selection. It explicitly combines "Smart Glasses," "CNN," and "Plant Identification," validating that the core concept of my project is technically feasible.

Support for Methodology: This paper influences my system architecture design. It demonstrates that wearable devices can successfully host Convolutional Neural Networks (CNNs). It supports my decision to focus on the integration between the camera module and the processing unit, assuring that the proposed hardware constraints are manageable for this specific application.

2. Kim, W.S. and Lee, S.K. (2020) 'Performance Evaluation of Plant Species Identification Using Deep Learning Models with Transfer Learning'. Journal of Management Information Systems.

Relevance to Project: This source is highly relevant as it focuses on "Performance Evaluation" and "Transfer Learning". It provides a direct academic precedent for my primary objective: comparing different models rather than building one from scratch.

Support for Methodology: This paper guides my "Model Selection" and "Evaluation" phases. It justifies the use of Transfer Learning (taking pre-trained models like MobileNet and retraining them on my specific dataset) as a standard, scientifically valid approach. It also helps define the metrics I must record—specifically accuracy and training time—to ensure my comparative analysis is rigorous.

3. Wang, J., Yang, Y., Wang, X., Xu, M., Wang, Y. and Wang, H. (2022) 'A Light-Weight Deep Learning Model for Plant Identification'. 2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC).

Relevance to Project: This paper is crucial for justifying the specific types of models I am comparing. It shifts the focus from general deep learning to "Light-Weight" models, which are essential for the limited battery and processing power of smart glasses.

Support for Methodology: This supports the "Model Optimisation" section of my methodology. It provides the theoretical backing for selecting architectures like MobileNet or EfficientNet over heavier models (like VGG16). It confirms that prioritizing low latency and small model size is a critical engineering requirement for wearable plant ID systems.

4. Sugi-Ganeshi, S. and Subramani, B. (2020) 'An Application of Deep Learning for Plant Identification based on Leaf Images on Smart Glasses'. 2020 5th International Conference on Devices, Circuits and Systems (ICDCS).

Relevance to Project: This source specifically targets "Leaf Images on Smart Glasses". It helps narrow the scope of my project from identifying whole plants (which is difficult with a head-mounted camera) to identifying specific features like leaves or flowers.

Support for Methodology: This directly supports my "Dataset Preparation" methodology. It justifies the curation of a dataset focused on macro-style images (leaves/flowers) that mimic the user's view through smart glasses, ensuring the training data matches the real-world deployment scenario.

5. Rzanny, M., Seeland, M. and Wäldchen, J. (2017) 'A mobile-based application for plant species identification using deep learning'. Multimedia Tools and Applications.

Relevance to Project: While this paper focuses on mobile phones, it is vital for establishing the constraints of mobile-based applications. Since smart glasses share similar hardware architectures (ARM processors, battery constraints) with mobile phones, this serves as a proxy for performance expectations.

Support for Methodology: This supports the "Comparative Analysis" phase. It highlights the trade-offs involved in running deep learning on portable devices. I will use the benchmarks established in this paper as a baseline to evaluate whether the performance I achieve on the smart glasses is within an acceptable range for a mobile robotic system.

2.4 Data Collection Review

Reflecting on the data collection plan and schedule:

The static image data collection phase is complete. The UK Leaf dataset was successfully downloaded and split into training (238 images), validation (68 images), and testing (34 images) subsets. The data augmentation pipeline was successfully applied, meaning no further static image collection is required at this stage .

Next Steps: The data collection focus now shifts from static training data to live testing data. Need to setup a raspberry pi to start testing and optimising the code.

Deliverables for review 3

- Review of Deliverables from Review 2
- Actions and record/logbook finished
- Skeleton report structure
- Abstract completed
- Raspberry Pi camera setup
- Models tested on Raspberry Pi

Review 3

3.1 Review of Deliverables from Review 2

1. Actions and Record/Logbook Finished

The project logbook has been fully updated. All action items from previous meetings have been documented, and the troubleshooting steps taken during the hardware configuration phase have been meticulously recorded to aid in writing the methodology section of the final report.

2. Skeleton Report Structure

A comprehensive table of contents has been developed. It breaks down the top-level narrative of the project, including bulleted lists of the key results and methodology for each section, ensuring a clear roadmap for the final submission on 31/03/2026.

3. Abstract Completed

The structured abstract has been drafted. It clearly articulates the project's core objectives, the significance of edge-based agricultural AI, and the successful outcomes of deploying the classification model onto the physical hardware.

4. Raspberry Pi Camera Setup

The physical camera hardware has been successfully integrated with the Raspberry Pi. This required overcoming significant software conflicts between the modern Raspberry Pi OS libcamera architecture and OpenCV's memory allocation. A live video feed has been successfully established and integrated into the Python environment.

5. Models Tested on Raspberry Pi

The trained classification model was successfully deployed to the physical edge device. Technical hurdles regarding TensorFlow version mismatches (specifically FULLY_CONNECTED version 12 optimization errors on the Pi's tflite-runtime library) were resolved by exporting an uncompressed .tflite model. The "Smart Glasses"

prototype can now successfully capture physical images and run real-time, on-device inferences.